Air Duct With Nozzles JR



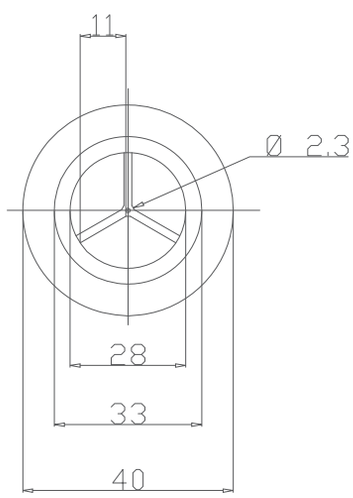
Description
Air duct with nozzles.

Application
Air supply

Characteristics
Available lenghts: 0.5m, 1.0m, 1.5m and 2.0m.
The nozzles are 32 mm in diameter, available in two colors: white RAL 9016 and black RAL 9005
The ideal temperature difference between supplied air and room air is between 2 and 4K and the velocity between 2.5 and 4.5 m/s.
Uniform and permanent air distribution.
Low pressure loss.

Materials and finishing
Galvanized steel sheet
Other RAL colors available on request.

Technical design



Selection table JR

Q (m³/h)	Length = 2000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
128 No. of rows 1	160	27	5.0
	200	26	5.0
	250	26	5.0
	315	26	5.0
	400	26	5.0
	500	26	5.0
249 No. of rows 2	160	34	7.0
	200	30	7.0
	250	29	7.0
	315	29	7.0
	400	29	7.0
	500	29	7.0
337 No. of rows 3	160	41	9.0
	200	32	9.0
	250	29	9.0
	315	29	9.0
	400	29	9.0
	500	29	9.0
413 No. of rowsi 4	160	>35	10.0
	200	35	10.0
	250	30	10.0
	315	29	10.0
	400	29	10.0
	500	29	10.0
483 No. of rows 5	160	>35	11.0
	200	>35	11.0
	250	31	11.0
	315	29	11.0
	400	29	11.0
	500	29	11.0
559 No. of rows 6	160	>35	12.0
	200	>35	12.0
	250	33	12.0
	315	29	12.0
	400	29	12.0
	500	29	12.0

Q (m³/h)	Length = 3000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
199 No. of rows 1	160	31	6.0
	200	29	6.0
	250	29	6.0
	315	29	6.0
	400	29	6.0
	500	29	6.0
381 No. of rows 2	160	-	8.0
	200	35	8.0
	250	32	8.0
	315	31	8.0
	400	31	8.0
	500	31	8.0
518 No. of rows 3	160	-	10.0
	200	40	10.0
	250	44	10.0
	315	31	10.0
	400	31	10.0
	500	31	10.0
637 No. of rows 4	160	-	11.0
	200	-	11.0
	250	36	11.0
	315	32	11.0
	400	31	11.0
	500	31	11.0
746 No. of rows 5	160	-	12.0
	200	-	12.0
	250	39	12.0
	315	32	12.0
	400	31	12.0
	500	31	12.0
859 No. of rows 6	160	-	13.0
	200	-	13.0
	250	-	13.0
	315	33	13.0
	400	31	13.0
	500	31	13.0

Q (m³/h)	Length= 4000 (mm)		
	D (mm)	NR [dB(A)]	ΔPt [pa]
270 No. of rows 1	160	36	9.0
	200	31	9.0
	250	30	9.0
	315	30	9.0
	400	30	9.0
	500	30	9.0
512 No. of rows 2	160	-	11.0
	200	40	11.0
	250	34	11.0
	315	33	11.0
	400	32	11.0
	500	32	11.0
698 No. of rows 3	160	-	16.0
	200	-	16.0
	250	38	16.0
	315	38	16.0
	400	32	16.0
	500	32	16.0
860 No. of rows 4	160	-	20.0
	200	-	20.0
	250	42	20.0
	315	33	20.0
	400	32	20.0
	500	32	20.0
1008 No. of rows 5	160	-	23.0
	200	-	23.0
	250	-	23.0
	315	36	23.0
	400	33	23.0
	500	32	23.0
1159 No. of rows 6	160	-	25.0
	200	-	25.0
	250	-	25.0
	315	38	25.0
	400	33	25.0
	500	32	25.0

Note

Described values are for:

- Velocity in sitting area <0.2 m/s (measured at 1.7 meters when mounting)
- Assembly height - 3m (coefficient value F1=1)
- Temperature difference - 3K-COOLING (coefficient value F2=1)

THE CALCULATED AIR FLOW WILL BE ADJUSTED BY MULTIPLYING WITH F1 AND F2 COEFFICIENTS, DEPENDING ON APPLICATION Qeff= Q *F1*F2.

Correction factor for temperature	
Temperature difference (K)	Correction coefficient F2
0	1.055
1	1.044
2	1.026
3	1
4	0.967
5	0.925
6	0.874

Correction factor for the mounting height	
Mounting height (m)	Correction coefficient F1
2.6	0.856
2.8	0.932
3	1
3.2	1.065
3.4	1.125
3.6	1.183
3.8	1.238
4	1.3

Annotation

- Dn (mm) - Duct diameter
- NR [dB(A)] - Noise level without the room attenuation
- ΔPt [pa] - Pressure loss

Order codes

Type	+	Ø	+	No. of rows	/	L	/	Nozzles color	/	0
JR		160 200 250 315 400 500		1 2 3 4 5 6		500 1000 1500 2000		WHITE RAL 9016 BLACK RAL 9005		Natural **0** Painted white RAL 9016 **R** Other RAL paints